Differentiation by first principles

1

a

a	b	$h = b - a$	$\frac{f(b) - f(a)}{b - a}$
3	4	1	37
3	3.5	0.5	31.75
3	3.1	0.1	27.91
3	3.05	0.05	27.4525
3	3.01	0.01	27.0901
3	3.001	0.001	27.009
3	3.0001	0.0001	27.0009

b

x	1	2	3	4	5
Instantaneous rate of change of $f(x)$ at x	3	12	27	48	75

c $3x^2$

2

a

a	b	$h = b - a$	$\frac{f(b) - f(a)}{b - a}$
3	4	1	7
3	3.5	0.5	6.5
3	3.1	0.1	6.1
3	3.05	0.05	6.05
3	3.01	0.01	6.01
3	3.001	0.001	6.001
3	3.0001	0.0001	6.0001

b

x	1	2	3	4	5
Instantaneous rate of change of $f(x)$ at x	2	4	6	8	10

c $2x$

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3

a

a	b	$h = b - a$	$\frac{f(b) - f(a)}{b - a}$
3	4	1	14
3	3.5	0.5	13
3	3.1	0.1	12.2
3	3.05	0.05	12.1
3	3.01	0.01	12.02
3	3.001	0.001	12.002
3	3.0001	0.0001	12.0002

b

x	1	2	3	4	5
Instantaneous rate of change of $f(x)$ at x	4	8	12	16	20

c $4x$

4

a

i $3x + 3h + 2$ **ii** $3h$ **iii** 3

b

i $8 - x - h$ **ii** $-h$ **iii** -1

c

i $2x + 2h - 3$ **ii** $2h$ **iii** 2

d

i $-5x - 5h + 4$ **ii** $-5h$

iii -5

e

i $5 - x^2 - 2hx - h^2$ **ii** $-2hx - h^2$

iii $-(2x + h)$

f

i $x^2 + 2hx + h^2 + 3x + 3h + 6$ **ii** $2hx + h^2 + 3h$

iii $2x + h + 3$

g

i $\frac{1}{x + h}$ **ii** $-\frac{h}{x(x + h)}$ **iii** $-\frac{1}{x(x + h)}$

h

i $\frac{1}{7(x + h)}$ **ii** $\frac{-h}{7x(x + h)}$ **iii** $-\frac{1}{7x(x + h)}$

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i

i $3x^2 + 6hx + 3h^2 - 4x - 4h + 2$

ii $6hx + 3h^2 - 4h$

iii $6x + 3h - 4$

5

a

i $f(x+h) = x^2 + 2hx + h^2$

ii $f(x+h) - f(x) = 2hx + h^2$

iii $\dfrac{f(x+h) - f(x)}{h} = 2x + h$

iv $\lim_{h \to 0} \left(\dfrac{f(x+h) - f(x)}{h} \right) = 2x$

b

i $f(x+h) = 4x^2 + 8hx + 4h^2$

ii $f(x+h) - f(x) = 8hx + 4h^2$

iii $\dfrac{f(x+h) - f(x)}{h} = 8x + 4h$

iv $\lim_{h \to 0} \left(\dfrac{f(x+h) - f(x)}{h} \right) = 8x$

c

i $f(x+h) = -6x^2 - 12hx - 6h^2$

ii $f(x+h) - f(x) = -12hx - 6h^2$

iii $\dfrac{f(x+h) - f(x)}{h} = -12x - 6h$

iv $\lim_{h \to 0} \left(\dfrac{f(x+h) - f(x)}{h} \right) = -12x$

6

a 0

b 0

c 5

d -2

e 2

f $2x$

g $8x$

h $2x - 3$

i $4x - 3$

j $8x - 3$

k $\dfrac{1}{2\sqrt{x}}$

l $\dfrac{1}{2\sqrt{x-5}}$

Gradients

7

a constant

b 4

c 4

8

a $2x + 8$

b 14

9

a $\dfrac{1}{2\sqrt{x}}$

b $\dfrac{1}{10}$

10

a $-2x$

b 2

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11

a $2x + 3$ **b** 11

12

a $3x^2$ **b** 12

13

a $4x^3$ **b** -500

14 24

15 144

16 -4

17 60

18

a $-\dfrac{1}{x^2}$ **b** $x = \pm 4$

19

a $-\dfrac{1}{2(x-2)^2}$ **b** $-\dfrac{1}{8}$

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